Attention to notation!

- Swap: Exchange rows.
  - $R_i = R_j$,
  - $R_i \leftrightarrow R_j$,
  - $r_i \leftrightarrow r_j$.
- Scaling: Multiply rows by constants.
  - cR_i,
  - $R_i = cR_i$,
  - cr_i.
- Combination: Add multiples of rows.
  - $R_i + cR_j$,
  - $R_i = R_i + cR_j$,
  - $R_i \leftarrow R_i + cR_j$.

After applying this row operations, systems of equations keep having the same solutions. If a system is augmented, the augmentation also is affected by the same row operations.

Remark: Given that systems are *equivalent* after row operations, we apply them in order to get simpler equations to solve.

Definition: A matrix is in <u>row echelon form</u> if:

1. The zero rows are at the **bottom**.
2. The 1st nonzero entry of each row **is to the right** of the 1st nonzero entry of the previous row.

Example: Consider the matrix

$$A = \begin{pmatrix} 1 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 3 \\ 0 & 0 & 2 & 4 & -5 \end{pmatrix}.$$

Let's see if A is in row echelon form.

C1 Condition 1 of the definition is met because there are no zero rows.

C2 The 1st nonzero entries of each row are 1, -1 and 2. The -1 in the second row is indeed to the **right** of the 1. But the 2 is to the **left** of the -1.

Condition 2 is not met, so A is not in row echelon form. But, can we convert it? Yes! With a row operation. Notice that by applying a **swap** we obtain:

$$\begin{pmatrix} 1 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & -1 & 3 \\ 0 & 0 & 2 & 4 & -5 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & 1 & -2 & 0 \\ 0 & 0 & 2 & 4 & -5 \\ 0 & 0 & 0 & -1 & 3 \end{pmatrix}.$$

The matrix A and the new matrix **are not the same**. However, this new matrix **is** in row echelon form.

Practice: Consider the matrix

$$B = \begin{pmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Is it in row echelon form? If not, convert it using row operations!

As a **hint**, you should use the **combination** operation.

Definition: A matrix is in <u>reduced row echelon form</u> (**RREF**) if:

1. It is in **row echelon form**.
2. The 1st nonzero entry of each row is a 1.
3. In each **column**, the leading 1s are **only accompanied by zeros**.

Example: Consider the last matrix we obtained from the previous example.

$$\begin{pmatrix} 1 & 0 & 1 & -2 & 0 \\ 0 & 0 & 2 & 4 & -5 \\ 0 & 0 & 0 & -1 & 3 \end{pmatrix}$$

Is it in **reduced row echelon form**?

C1 Yes, it is in row echelon form.

C2 The 1st nonzero entries of each row are 1, 2 and -1. So only the first row meets this condition.

We remedy this by applying two **scalings**:

$$\begin{pmatrix} 1 & 0 & 1 & -2 & 0 \\ 0 & 0 & 2 & 4 & -5 \\ 0 & 0 & 0 & -1 & 3 \end{pmatrix} \xrightarrow{\frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 2 & \frac{-5}{2} \\ 0 & 0 & 0 & -1 & 3 \end{pmatrix}$$

$$\xrightarrow{-1R_3} \begin{pmatrix} 1 & 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 2 & \frac{-5}{2} \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix}$$

With this, condition 2 is now met. However, condition 3 is still not met. The 1 in the second row has a 1 above it, and the one in the third row has a -2 and a 2. To get rid of them, we apply a **combination**.

$$\begin{pmatrix} 1 & 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 2 & \frac{-5}{2} \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix} \xrightarrow{R_1-R_2} \begin{pmatrix} 1 & 0 & 0 & -4 & \frac{5}{2} \\ 0 & 0 & 1 & 2 & \frac{-5}{2} \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix}$$

$$\xrightarrow{R_2-2R_3} \begin{pmatrix} 1 & 0 & 0 & -4 & \frac{5}{2} \\ 0 & 0 & 1 & 0 & \frac{7}{2} \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix}$$

$$\xrightarrow{R_1+4R_3} \begin{pmatrix} 1 & 0 & 0 & 0 & \frac{-19}{2} \\ 0 & 0 & 1 & 0 & \frac{7}{2} \\ 0 & 0 & 0 & 1 & -3 \end{pmatrix}$$

Condition 3 is thus met, and this is the reduced row echelon form of A.

Remark: By saying *the* reduced row echelon form, we allude to uniqueness. This is because **every matrix** is row-equivalent to a **unique** matrix in **RREF**.

Practice: Let's solve the following system of equations in variables (a,b,c,d):

$$\begin{cases} 3a-c=0 \\ 8a-2d=0 \\ 2b-2c-d=0 \end{cases}$$

Practice: Consider the system of equations in variables (u,v,w,x,y) below:

$$\begin{cases} u+v-y=1 \\ v+2w+x+3y=1 \\ u-w+x+y=0 \end{cases}$$

1. What is the coefficient matrix associated with the system? And the augmented matrix?
2. Reduce the augmented matrix to find the solution. ?
3. Which variables are free in the solution? How many parameters does the solution depend on?

The Matrix Product

The condition that guarantees that two matrices A,B can be multiplied is: If A is $[m\times p]$ and B is $[p\times n]$, then the entries of the product are defined by the formula: More simply: The matrix AB will be of size $[m\times n]$.

Example: Consider the matrices

$$A=\begin{pmatrix} 1 & -1 \\ 3 & 4 \\ -2 & 0 \\ 0 & 4 \end{pmatrix}, B=\begin{pmatrix} 2 & 0 & -2 & 8 & -5 \\ 1 & -2 & 3 & 9 & 6 \end{pmatrix}.$$

Notice that:

- A is $[4\times 2]$.
- B is $[2\times 5]$.
  - It holds that #cols. A=#rows B.
  - We can multiply A and B!
- The matrix AB will be of size $[4\times 5]$.

If we want to find the $(2,4)$ entry of AB, we use row 2 of A and column 4 of B:

$$\begin{pmatrix} 1 & -1 \\ 3 & 4 \\ -2 & 0 \\ 0 & 4 \end{pmatrix} \quad \text{y} \quad \begin{pmatrix} 2 & 0 & -2 & 8 & -5 \\ 1 & -2 & 3 & 9 & 6 \end{pmatrix}.$$

We take the dot product of these vectors:

$$(3,4)\cdot(8,9)=3\cdot 8+4\cdot 9=60$$

and therefore $(AB)_{24}=60$.

In general, we can arrange the matrices in the following way:

		2	0	−2	8	−5
		1	−2	3	9	6
1	−1					
3	4				(∗)	
−2	0					
0	4					

Practice:

- Find the $(1,3)$ entry of AB by highlighting the corresponding rows and columns.
- Perform the same procedure to find the $(3,2)$ entry of AB.

To find any entry, we locate the corresponding row and column of the matrices and calculate their dot product.

Exercise: We already multiplied A on the left of B. Can we multiply BA? That is, reversing the order of multiplication.

Remark: In general, if A has dimensions $[m\times n]$ and B has dimensions $[n\times m]$, then both AB and BA exist.

Example: Consider the matrices

$$A=\begin{pmatrix} 1 & -1 \\ 3 & 4 \\ -2 & 0 \\ 0 & 4 \end{pmatrix}, B=\begin{pmatrix} 2 & 0 & -2 & 8 & -5 \\ 1 & -2 & 3 & 9 & 6 \end{pmatrix}.$$

If we want to find AB, we first verify if we can multiply them. A is $[2\times 4]$ and B is $[4\times 2]$; in this case,

$$\text{\#rows } A=\text{\#cols. } B$$

so we can calculate AB.

In the same way, if we look for BA, we can also calculate it because

$$\#\text{rows } B = \#\text{cols. } A.$$

Practice:

1. What size will the matrices AB and BA have?
2. Work together with someone to find the resulting matrices from the multiplication.

Determinants

Notation: The determinant of a matrix A is denoted by $\det(A)$ or $|A|$.

- In the $[2\times 2]$ case, if $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then
$$\det(A) = ad - bc.$$
- The $[3\times 3]$ case, **Sarrus' Rule**:
If $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, then we calculate its determinant by taking the first two columns and augmenting A with them such that we obtain
$$\left(\begin{array}{ccc|cc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{array}\right)$$
from here we add the diagonals from top to bottom and subtract the diagonals from bottom to top such that
$$\det(A) = aei + bfg + cdh - gec - hfa - idb.$$

Eigenvalues and Eigenvectors

Eigenvectors are generators for invariant spaces of linear maps; that is, *they don't change direction after the mapping.*

Definition: An eigenvector of a map T is a non-zero vector $\vec{x}$ that satisfies $T(\vec{x}) = \lambda\vec{x}$ for some scalar λ. The constant λ is called the eigenvalue associated with $\vec{x}$.

To find the eigenvectors, we modify the defining equation $T(\vec{x}) = \lambda\vec{x}$ as follows:

$$\begin{aligned} T(\vec{x}) = \lambda\vec{x} &\iff T(\vec{x}) = \lambda \mathrm{id}(\vec{x}) \\ &\iff T(\vec{x}) - \lambda \mathrm{id}(\vec{x}) = 0 \\ &\iff (T - \lambda \mathrm{id})(\vec{x}) = 0 \\ &\iff (A - \lambda I)(\vec{x}) = 0 \end{aligned}$$

So we are tasked with finding the solutions $\vec{x}$ to the matrix equation:

$$(A - \lambda I)(\vec{x}) = 0.$$

Remark: For the matrix in question to have non-zero solutions to the homogeneous equation, we must guarantee that it is non-invertible. So that:

$$A - \lambda I \text{ is non-invertible} \iff \det(A - \lambda I) = 0.$$

Definition: For a linear map T represented by a matrix A, we define its characteristic polynomial as:

$$f_A(x) = \det(A - xI).$$

The zeroes, or roots, of the characteristic polynomial will be T's eigenvalues.

Remark: Observe that geometrically, we often start by identifying the eigenvectors, because invariant subspaces can be visually perceived. However, algebraically we proceed in reverse: finding the eigenvalues first and then their corresponding eigenvectors.

Definition: The algebraic multiplicity (a.m.) of an eigenvalue λ is the degree of the factor $(x - \lambda)$ in the characteristic polynomial. It is also the number of times λ appears as an eigenvalue of the map.

The geometric multiplicity (g.m.) of an eigenvalue λ is the number of linearly independent eigenvectors associated with λ.

Important: It always holds that:

$$1 \leq \text{g.m.} \leq \text{a.m.}$$

In other words: every eigenvalue always has at least one eigenvector associated with it. Furthermore, there can be no more linearly independent eigenvectors than instances of the eigenvalue they are associated with.

Example: We have $C = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$, a shear mapping.

- C is triangular $\Rightarrow \lambda = 1$ with a.m.$=2$.
- It has a single eigenvector $(1,0)$, so g.m.$=1$.
- Since a.m. $\neq$ g.m., C is not diagonalizable.

Example: Let's consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

We are looking for its eigenvectors.

1. Using the trace-determinant formula we have that:
$$m = \frac{1}{2}(2+2),\ p = \det A = 3 \Rightarrow \lambda = 2 \pm \sqrt{2^2 - 3},$$
so $\lambda_1 = 3$ and $\lambda_2 = 1$.
2. The first eigenspace is:
$$E_{\lambda_1} = \ker(A - 3I) = \ker\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} = \ker\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}.$$

The equation associated with this matrix is $x-y=0$ and its solution is:

$$(x,y)=(x,x)=x(1,1)\Rightarrow E_{\lambda_1}=\text{span}(1,1).$$

We conclude that $\vec{v}_1=(1,1)$ is an eigenvector of A associated with $\lambda_1=3$.

3. The second eigenspace is:

$$E_{\lambda_2}=\ker(A-I)=\ker\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}=\ker\begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix}.$$

The associated equation is $x+y=0$. Its solution is:

$$(x,y)=(x,-x)=x(1,-1)\Rightarrow E_{\lambda_2}=\text{span}(1,-1).$$

and thus the other eigenvector is $\vec{v}_2=(1,-1)$ associated with $\lambda_2=1$.

Example: Let's consider

$$A=\begin{pmatrix}0 & 0 & -2\\ 0 & 2 & 2\\ -2 & 2 & 1\end{pmatrix}.$$

We are looking for its eigenvectors.

1. To find the eigenvalues, we must find the roots of the characteristic polynomial. We compute the determinant:

$$f(\lambda)=\det(A-\lambda I)=\det\begin{pmatrix}-\lambda & 0 & -2\\ 0 & 2-\lambda & 2\\ -2 & 2 & 1-\lambda\end{pmatrix}.$$

To expand this determinant, we can use Sarrus' rule. We augment the matrix with its first two columns:

$$\left(\begin{array}{ccc|cc}-\lambda & 0 & -2 & -\lambda & 0\\ 0 & 2-\lambda & 2 & 0 & 2-\lambda\\ -2 & 2 & 1-\lambda & -2 & 2\end{array}\right)$$

Next, we sum the products of the top-to-bottom diagonals and subtract the products of the bottom-to-top diagonals:

$$\begin{aligned}f(\lambda)&=[(-\lambda)(2-\lambda)(1-\lambda)+(0)(2)(-2)+(-2)(0)(2)]\\&\quad-[(-2)(2-\lambda)(-2)+(2)(2)(-\lambda)+(1-\lambda)(0)(0)]\\&=[-\lambda(2-3\lambda+\lambda^2)+0+0]-[4(2-\lambda)-4\lambda+0]\\&=(-\lambda^3+3\lambda^2-2\lambda)-(8-4\lambda-4\lambda)\\&=(-\lambda^3+3\lambda^2-2\lambda)-(8-8\lambda)\\&=-\lambda^3+3\lambda^2+6\lambda-8.\end{aligned}$$

By factoring the polynomial, we obtain $f(\lambda)=-(\lambda-1)(\lambda-4)(\lambda+2)$, so our eigenvalues are $\lambda_1=1$, $\lambda_2=4$, and $\lambda_3=-2$. Because all three eigenvalues are distinct, we already know that a.m.$=1$ for each of them.

2. The first eigenspace is:

$$E_{\lambda_1}=\ker(A-I)=\ker\begin{pmatrix}-1 & 0 & -2\\ 0 & 1 & 2\\ -2 & 2 & 0\end{pmatrix}.$$

We find the reduced row echelon form (RREF) of this matrix:

$$\begin{pmatrix}1 & 0 & 2\\ 0 & 1 & 2\\ 0 & 0 & 0\end{pmatrix}$$

This yields the equations $x+2z=0$ and $y+2z=0$. Its solution is:

$$\begin{aligned}(x,y,z)&=(-2z,-2z,z)\\&=z(-2,-2,1)\\\Rightarrow E_{\lambda_1}&=\text{span}(-2,-2,1).\end{aligned}$$

We conclude that $\vec{v}_1=(-2,-2,1)$ is an eigenvector of A associated with $\lambda_1=1$.

3. The second eigenspace is:

$$E_{\lambda_2}=\ker(A-4I)=\ker\begin{pmatrix}-4 & 0 & -2\\ 0 & -2 & 2\\ -2 & 2 & -3\end{pmatrix}.$$

Finding its RREF gives:

$$\begin{pmatrix}1 & 0 & 1/2\\ 0 & 1 & -1\\ 0 & 0 & 0\end{pmatrix}$$

This yields the equations $x+\frac{1}{2}z=0$ and $y-z=0$. To avoid fractions in our vector, we can set $z=-2x$, which then gives $y=-2x$. Its solution is:

$$\begin{aligned}(x,y,z)&=(x,-2x,-2x)\\&=x(1,-2,-2)\\\Rightarrow E_{\lambda_2}&=\text{span}(1,-2,-2).\end{aligned}$$

Thus, the second eigenvector is $\vec{v}_2=(1,-2,-2)$ associated with $\lambda_2=4$.

4. The third eigenspace is:

$$E_{\lambda_3}=\ker(A+2I)=\ker\begin{pmatrix}2 & 0 & -2\\ 0 & 4 & 2\\ -2 & 2 & 3\end{pmatrix}.$$

Finding its RREF gives:

$$\begin{pmatrix}1 & 0 & -1\\ 0 & 1 & 1/2\\ 0 & 0 & 0\end{pmatrix}$$

This yields the equations $x-z=0$ and $y+\frac{1}{2}z=0$. To avoid fractions, we set $z=-2y$ and thus $x=-2y$. Its solution is:

$$\begin{aligned}(x,y,z)&=(-2y,y,-2y)\\&=y(-2,1,-2)\\\Rightarrow E_{\lambda_3}&=\text{span}(-2,1,-2).\end{aligned}$$

And thus the final eigenvector is $\vec{v}_3=(-2,1,-2)$ associated with $\lambda_3=-2$.

We see that for all three eigenvalues, a.m.=g.m.$=1$.

Generalized eigenvectors

The geometric idea of a generalized eigenspace is that it is an entire line, plane, or subspace *which doesn't pass through the origin* (*it is an affine space!*) that, when acted upon by a linear map, gets pushed into a standard eigenspace.

Definition: If $\vec{v}_1$ is a standard eigenvector associated with λ, a generalized eigenvector $\vec{v}_2$ is a vector that gets pushed into the direction of $\vec{v}_1$. Algebraically, we find it by solving the non-homogeneous matrix equation:

$$(A-\lambda I)\vec{v}_2=\vec{v}_1$$

Example: We have the shear mapping from before,

$$C=\begin{pmatrix}1 & 3\\ 0 & 1\end{pmatrix},$$

we know:

- C is triangular $\Rightarrow \lambda=1$ with a.m.$=2$.
- It has a single eigenvector $\vec{v}_1=(1,0)$, so g.m.$=1$.
- Since a.m. $\neq$ g.m., C has a generalized eigenvectors.

To find the generalized eigenvector $\vec{v}_2$, we solve the equation $(C-I)\vec{v}_2=\vec{v}_1$:

$$\begin{pmatrix}0 & 3\\ 0 & 0\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}1\\ 0\end{pmatrix}$$

This yields the equation

$$3y=1\Rightarrow y=1/3$$

and x is free. So that the solution to the system is

$$(x,y)=(x,1/3)=x(1,0)+(0,1/3).$$

The solution includes the eigenvector itself but also the generalized eigenvector which we have not seen before. Thus, $\vec{v}_2=(0,1/3)$ is a generalized eigenvector for C.

Example: Let's consider $A=\begin{pmatrix}0 & -1 & -1\\ 1 & 2 & 0\\ 0 & 0 & 2\end{pmatrix}$. We will find all its eigenvectors and generalized eigenvectors.

1. First, we find the characteristic polynomial:

$$f(\lambda)=\det(A-\lambda I)=\det\begin{pmatrix}-\lambda & -1 & -1\\ 1 & 2-\lambda & 0\\ 0 & 0 & 2-\lambda\end{pmatrix}.$$

Expanding along the third row gives:

$$\begin{aligned}f(\lambda)&=(2-\lambda)\det\begin{pmatrix}-\lambda & -1\\ 1 & 2-\lambda\end{pmatrix}\\ &=(2-\lambda)(-\lambda(2-\lambda)-(-1))\\ &=(2-\lambda)(\lambda^2-2\lambda+1)\\ &=(2-\lambda)(\lambda-1)^2.\end{aligned}$$

Our eigenvalues are $\lambda_1=1$ (with a.m.$=2$) and $\lambda_2=2$ (with a.m.$=1$).

2. Let's find the eigenspace for $\lambda_1=1$:

$$E_{\lambda_1}=\ker(A-I)=\ker\begin{pmatrix}-1 & -1 & -1\\ 1 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}.$$

We find the RREF of this matrix:

$$\begin{pmatrix}1 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{pmatrix}$$

This yields the equations $x+y=0$ and $z=0$. Its solution is:

$$\begin{aligned}(x,y,z)&=(-y,y,0)\\ &=y(-1,1,0)\\ \Rightarrow E_{\lambda_1}&=\text{span}(-1,1,0).\end{aligned}$$

We conclude that $\vec{v}_1=(-1,1,0)$ is the only regular eigenvector for $\lambda_1=1$. Since g.m.$=1$ but a.m.$=2$, we need a generalized eigenvector!

3. To find the generalized eigenvector $\vec{v}_2$ associated with $\lambda_1=1$, we solve $(A-I)\vec{v}_2=\vec{v}_1$. We set up the augmented matrix:

$$\left(\begin{array}{ccc|c}-1 & -1 & -1 & -1\\ 1 & 1 & 0 & 1\\ 0 & 0 & 1 & 0\end{array}\right)$$

Finding its RREF gives:

$$\left(\begin{array}{ccc|c}1 & 1 & 0 & 1\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0\end{array}\right)$$

This yields the equations $x+y=1$ and $z=0$. From here

$$y=1-x$$

so that our solution looks like

$$\begin{aligned}(x,y,z)&=(x,1-x,0)\\ &=(-x)(-1,1,0)+(0,1,0)\end{aligned}$$

Thus, $\vec{v}_2=(0,1,0)$ is our generalized eigenvector and we see the original eigenvector in the solution.

4. Finally, we find the eigenspace for $\lambda_2 = 2$:

$$E_{\lambda_2} = \ker(A - 2I) = \ker \begin{pmatrix} -2 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Finding its RREF gives:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

This yields the equations $x = 0$ and $y + z = 0$. Its solution is:

$$\begin{aligned} (x,y,z) &= (0,-z,z) \\ &= z(0,-1,1) \\ \Rightarrow E_{\lambda_2} &= \operatorname{span}(0,-1,1). \end{aligned}$$

And thus the third regular eigenvector is $\vec{v}_3 = (0,-1,1)$ associated with $\lambda_2 = 2$.